

Jalal Ostadhadi Dehkordi

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AI Engineer with experience in the oil and gas industry and research in machine learning and deep learning. Passionate about developing practical AI solutions across various domains. Experienced in teaching, mentoring hands-on projects, and conducting applied research in both academic and industrial settings.

Skills

- **Machine Learning**: Supervised and Unsupervised learning, Feature engineering, SVM, Ensemble methods
- **Deep Learning**: CNN architectures, Transfer learning, Fine-tuning, GANs
- **NLP**: RAG, Prompt engineering, Sentiment analysis, Text classification
- **Computer Vision**: Image preprocessing, Filtering, Augmentation, Object detection
- **Model Optimization:** Hyper parameter tuning, Cross-Validation, Precision/Recall evaluation
- **Data Science**: Data wrangling, Visualization, EDA
- **Data Mining** : Apriori, FP-Growth, Association Rule Mining
- Automation & Workflow Design: N8N Agent Design and Automation
- **Database Management** : SQL, MySQL

Experience

2025 – Present | AI Department Manager | Medeed

- Lead the AI department, building automation that extracts structured data from PDFdocuments in German, English, and Persian.

2024 – [Present] | AI Instructor | University of Applied Science and Technology

- Teaching IT and AI courses.

2023 – 2025 | Research Assistant | Deep Learning Lab, University of Tehran

- Collaborated on deep learning research projects and developed experimental models.

2023 – 2024 | AI Researcher, Oil Industry | RAMPCO International Company

- Conducted training sessions and introduced practical AI projects for the oil industry.

2022 – 2023 | Artificial Intelligence Researcher | PetroPala (Part-time)

- Research on ML/DL for predictive maintenance using IIoT sensor data.

2021 – 2022 | ASP.Net Developer | Arad Sanat Gharb, Iran

- Developed user interfaces using HTML, CSS, JavaScript, and Bootstrap for web applications.

Education

✪ **MS**	University of Tehran,	**IT Artificial Intelligence**	GPA : 18.30
✪ **BS**	University of Applied Science,	**IT Web Programming**	GPA : 18.46
✪ **AAS**	University of Applied Science	**Software Engineering**	GPA : 18.50

Master's Thesis:

Weld Defect Classification from Industrial Radiography Images Using Deep Learning

Developed and trained multiple deep learning models using CNN and ViT architectures to classify weld defects with Python, TensorFlow, PyTorch, LaTeX, Mendeley

Practical Training & Certifications:

Completed multiple hands-on courses on Kaggle to strengthen practical skills in data science and AI

Programming	Data Science & Analysis	Artificial Intelligence
• Python	• Data Cleaning	• Intro to Machine Learning
• Intro to Programming	• Pandas	• Intermediate Machine Learning
Time series analysis	• Feature Engineering	• Intro to Deep Learning
• Intro to Time Series	• Data Visualization	• Computer Vision

Conferences & Academic Activities

- **Workshop Instructor** at the 7th International Conference on Cyberspace (CYSP-2024), , on Applications of Artificial Intelligence in Inspection Industries, University of Tehran.
- **Session Secretary** at two consecutive editions of the International Conference on Cyberspace (CYSP-2022 & CYSP-2023), University of Tehran.
- **Session Secretary** at the 6th International Conference on Pattern Recognition and Image Analysis (IPRIA-2023), University of Tehran.
- **Research Assistant** at the Deep Learning Research Laboratory and Cyberspace Laboratory, University of Tehran.

Publications (Scientific Papers)

- **Conference Paper:** Weld defect classification in industrial radiographic images using a proposed deep learning model, presented at the International Conference on Pattern Recognition and Image Analysis (IPRIA-2023), indexed in the CIVILICA database.
- **Journal Paper:** Introduction of the deep learning model WeldClassNet for weld defect classification from industrial radiographic images, submitted to the Journal of *Machine Vision and Image Processing***.**
- **Journal Paper:** Weld defect classification from X-Ray images using three lightweight deep ensemble learning techniques, submitted to the (*AUT) Journal of Electrical Engineering.*
- **Journal Paper:** A unified machine learning and deep learning framework for multiclass predictive maintenance (PdM) using temporal sensor data in IIoT systems, submitted to the *Iranian Journal of Industrial Engineering***.**

Book Translation:

Book Original: Countering Cyberterrorism: The Confluence of Artificial Intelligence, Digital Policing, and Cyber Forensics in US and UK National Cybersecurity

Publication Format: Electronic **Publication Year:** 2024 **ISBN:** 978-3-031-21920-7

Activities & Projects

Scientific & Applied Projects : (Implemented as practical projects and educational scripts on GitHub and Kaggle)

DATA SCIENCE & ANALYSIS

- **Data Wrangling (Simulated Dataset):** Data cleaning, handling missing values, and preparing datasets for analysis.
- **Visual Data Analysis (Telecom Churn):** Data analysis and visualization using Python and libraries such as Pandas, Matplotlib, and Seaborn to explore data distributions and hidden patterns.

- **Dimensionality Reduction with PCA (Sklearn – Breast Cancer Dataset):** Applied PCA for dimensionality reduction and plotted correlation heatmaps to analyze relationships between features and principal components.
- **SQL with BigQuery (Kaggle):** Analysis of public datasets (e.g., Hacker News) using SQL and BigQuery.

MACHINE LEARNING

- **K-Means Clustering (Titanic Dataset):** clustering unlabeled data to discover patterns, with visualization and analysis of clustering results.
- **Binary Classification Problems (Titanic Dataset):** Implementation and comparison of Logistic Regression, Naïve Bayes, and SVM models.
- **Regression Models (USA Housing Dataset):** Training and evaluation of multiple regression models for prediction tasks.
- **Classification Models (Classified_Data.csv):** Training and evaluation of KNN, Decision Tree (DT), and Random Forest (RF) models.
- **SVM Classification (Iris Dataset):** Training an SVM model and optimizing it using GridSearchCV.

DEEP LEARNING

- **WeldClassNet (Private Dataset):** Design and implementation of a deep learning model for weld defect detection and classification using the EfficientNetB1 architecture.
- **DCGAN – MNIST:** Implementation of DCGAN using TensorFlow/Keras for handwritten digit generation.
- **Binary Text Classification (IMDB):** Movie review sentiment classification (positive/negative) using deep neural networks in Keras.
- **Image Classification (Dogs vs Cats):** CNN model design using Regularization, Dropout, and Data Augmentation techniques.
- **Regression with Keras (Boston Housing):** House price prediction using data normalization and K-Fold cross-validation.

COMPUTER VISION & IMAGE PROCESSING

- **Image Processing – Welding X-Ray:** Processing industrial X-Ray images using Laplacian, Gaussian, Sobel, and Canny filters for feature extraction and image enhancement.
- **Defect Classification in Welding X-Ray (RIAWELC):** Weld region segmentation, image quality enhancement, and implementation of multiple deep learning models for weld defect classification.

INTELLIGENT AND INDUSTRIAL SYSTEMS

- **Water Quality Intelligent Monitoring System:** Design of an intelligent water quality monitoring system using ANN and CNN models to predict water safety for aquatic life.
- **Predictive Maintenance (PdM) Project – IIoT:** Design and implementation of a multiclass predictive maintenance system using temporal sensor data, including feature engineering and machine learning/deep learning models for industrial fault prediction.

GENERATIVE AI, NLP, AND INTELLIGENT AGENTS

- **Gemini API Experiments:** Practical guide for working with Google Gemini models on Kaggle, including API configuration, content generation, conversation, and advanced prompt engineering techniques such as Zero-shot, Few-shot learning, Chain-of-Thought, and Reason + Act.
- **Gemini API Educational Project:** Educational guide for using Google Gemini API, covering Text Embeddings, Similarity Scoring, RAG-based Q&A, and Text Classification with Keras.

- **Conversational AI Systems:** Two educational chatbot projects: *BaristaBot*(Cafe order-taking chatbot) and *Database Chatbot* (Conversational interface for database querying and SQL execution using Gemini API)
- **Gemini API Search Grounding Guide:** Guide to using Search Grounding in Gemini API for retrieving verified and up-to-date web information via Google Search, including setup in AI Studio.
- **Fine-Tuned Gemini Text Classification:** Educational project on text classification using the Newsgroups20 dataset, including data preparation, model fine-tuning, and performance evaluation with Google Gemini.

AGENTS & AUTOMATION

- **AI Agent Design with n8n:** Design and implementation of intelligent agents using n8n for workflow automation.
- **Telegram AI Bot (n8n-based):** Development of a Telegram bot using n8n and AI APIs for user interaction, request processing, and intelligent responses.

Interests :

- **Data Science & Intelligent Analysis**: Finding hidden patterns and insights in data
- **Developing and optimizing machine learning and deep learning models** for real-world applications
- **Natural Language Processing (NLP):** Information extraction, sentiment analysis, and building intelligent chat systems
- **Research & Innovation**: Exploring the latest advances and emerging technologies in AI and deep learning
- **Teaching & Knowledge Sharing**: Educating others in AI concepts and inspiring the next generation